

Named Entities & Responsible NLP

Post-Assessment 3 · 9 questions · 33 points

Name: _____

Date: _____

Score: / 33

This is a **post-assessment**. It shows what you learned in this unit. Try every question, and check your answers if you have time.

Work on your own, without a computer. You have about **40 minutes**. When a question asks what a program prints, write exactly what Python would print, one line of output per answer line. If the code would crash, write the name of the error (for example, `TypeError`). Show your thinking; partial credit is given.

In every spaCy question, assume this code has already run: `import spacy` and `nlp = spacy.load("en_core_web_sm")`

1. Name that entity

5 points

Write the spaCy label for each entity: **PERSON**, **ORG**, **GPE**, **DATE**, or **MONEY**. Each label is used once.

Entity	Label
\$20	
Tokyo	
Microsoft	
last Tuesday	
Albert Einstein	

2. Entity finder

4 points

```
doc = nlp("Omar moved to Canada in 2019 with his mom.")
for ent in doc.ents:
    print(ent.text, ent.label_, ent.start_char)
```

(a) Write what the program prints. (spaCy finds three entities here.)

(3 pts)

(b) What does `start_char` tell you?

(1 pt)

3. Entity Gallery

2 points

(a) What is the difference between `style="ent"` and `style="dep"` in `displaCy`? (1 pt)

(b) Why might you save a `displaCy` picture as an HTML file? (1 pt)

4. When the model is wrong

4 points

Here is what `spaCy` really found in two sentences.

SPACY OUTPUT

Jordan scored 30 points.	→ Jordan GPE, 30 CARDINAL
we flew to paris with amara.	→ paris GPE

(a) In the first sentence, what should the label for “Jordan” be, and why did the model get it wrong? (2 pts)

(b) In the second sentence, which entity did the model **miss**, and why is this sentence harder? (2 pts)

5. NER Report Card

3 points

These are real spaCy guesses. For each row, write ✓ if the label is right or ✗ if it is wrong. For each ✗, write the correct label (or “none” if it isn't an entity).

Sentence	Entity	Model's label	✓ or ✗	Correct label if ✗
Leo moved to Canada in 2019.	Leo	ORG		
Leo moved to Canada in 2019.	Canada	GPE		
Jordan scored 30 points.	Jordan	GPE		
Yesterday Leo Martinez called me.	Leo Martinez	PERSON		
Leo moved to Canada in 2019.	2019	DATE		

(a) Complete the ✓/✗ column and the corrections (in the table above). (2 pts)

(b) What is the model's score, as a fraction and as a percent? (1 pt)

6. Character Tracker

4 points

```
from collections import Counter
names = ["Alice", "White Rabbit", "Alice Liddell", "Queen", "Alice", "Rabbit"]
aliases = {"Alice Liddell": "Alice", "White Rabbit": "Rabbit"}
clean = [aliases.get(n, n) for n in names]
counts = Counter(clean)
print(counts.most_common(2))
```

(a) What does it print? (2 pts)

(b) Without the `aliases` step, what would `Counter(names)["Rabbit"]` be? (1 pt)

(c) Write a list comprehension that collects the text of every **GPE** entity in `doc`. (1 pt)

7. How models learn

3 points

(a) What is **training data**?

(1 pt)

(b) Suppose a model was trained mostly on neat, correctly capitalized articles. How might it do on kids' text messages? Why?

(1 pt)

(c) Name one way to check whether a model treats different people's names fairly.

(1 pt)

8. Responsible NLP

3 points

A company wants to scan kids' chat messages to find names and places.

(a) Give one possible **benefit**.

(1 pt)

(b) Give one **privacy or fairness** risk.

(1 pt)

(c) Write one sentence for this tool's **Limitations** card.

(1 pt)

9. Place Counter

5 points

Write a function `places_count(text)` that runs `nlp` on the text and **returns** a `Counter` of how many times each GPE (place) entity appears.

EXAMPLE: `PLACES_COUNT("WE DROVE FROM CHICAGO TO DENVER, THEN BACK TO CHICAGO.")` RETURNS

```
Counter({'Chicago': 2, 'Denver': 1})
```


END OF ASSESSMENT